MAILTRACE AI // FORENSIC TELEMETRY PLATFORM

# Digital Evidence & Forensic Incident Dossier

| | | | |
|---|---|---|---|
| **REPORT REFERENCE:** | RPT-■20260916-■3736E7 | **GENERATED (UTC):** | 2026-■09-■16 21:■05:■47 UTC |
| **EVIDENCE IDENTIFIER:** | sample-■001 | **ASSIGNED CASE:** | UNASSIGNED /■ DIRECT EVIDENCE |
| **INVESTIGATOR:** | SOC Lead Analyst | **CLASSIFICATION:** | **CONFIDENTIAL // EVIDENTIARY DISCLOSURE** |
| **EMAIL SHA-256:** | `97d4b2e811c7520e5e79603f9050■d268159b360b9432df03d4083d8e■57ef228a` | **HASH ALGORITHM:** | FIPS PUB 180-4 SHA-256 (Preserved Literal) |

## 1. Executive Forensic Summary

| DETERMINISTIC THREAT SCORE | ASSESSED SEVERITY | ML PHISHING PROBABILITY | ML CLASSIFICATION |
|---|---|---|---|
| **82/100** | **CRITICAL** | **89.0%** | **PHISHING** |

Email risk assessed as Critical (82/100) driven by 8 forensic signal(s): Brand impersonation (+18), Credential harvesting (+12), DMARC failure (+12), Link mismatch (+15), NLP phishing classifier (+10).

## 2. Email Headers & Sender Alignment

| | |
|---|---|
| **From:** | `Microsoft Security &■lt;■security@■micros0ft-■example.■com&■gt;■` |
| **To:** | `employee@■company.■com` |
| **Subject:** | **URGENT:■ Verify your Microsoft Account** |
| **Date:** | Sun,■ 7 Sep 2026 10:■42:■00 +■0530 |
| **Reply-To:** | `support@■example.■net` |
| **Return-Path:** | `bounce@■example.■net` |
| **Message-ID:** | `&■lt;■abc123@■example.■com&■gt;■` |

| | | | |
|---|---|---|---|
| **From Domain:** | `micros0ft-■example.■com` | **Reply-To Domain:** | **example.■net (MISMATCH - SUSPICIOUS)** |
| **Return-Path Domain:** | **example.■net (MISMATCH)** | **Alignment Verdict:** | **REPLY-TO / SENDER SPOOFING DETECTED** |

## 3. Email Authentication Findings (SPF / DKIM / DMARC)

| PROTOCOL | OBSERVED VERDICT | TECHNICAL DETAILS & EVIDENCE |
|---|---|---|
| **SPF (Sender Policy)** | **FAIL** | spf=■fail (domain of bounce@■example.■net does not designate 203.■0.■113.■25) |
| **DKIM (Cryptographic)** | **FAIL** | dkim=■fail header.■i=■@■company.■com |
| **DMARC (Domain Policy)** | **FAIL** | dmarc=■fail (p=■reject dis=■none) |

*Notice: Observed authentication result from supplied headers (unverified by local mail server)*

## 4. Deterministic Threat Score Breakdown

| CATEGORY | IMPACT | FORENSIC FINDING & JUSTIFICATION |
|---|---|---|

| | | |
|---|---|---|
| **General** | **+18 pts** | N/A |
| **General** | **+15 pts** | N/A |
| **General** | **+12 pts** | N/A |
| **General** | **+12 pts** | N/A |
| **General** | **+10 pts** | N/A |
| **General** | **+10 pts** | N/A |
| **General** | **+8 pts** | N/A |
| **General** | **+8 pts** | N/A |

## 5. NLP Content Phishing Classification

| | | | |
|---|---|---|---|
| **Model Name:** | TF-■IDF +■ Logistic Regression | **Classification:** | **PHISHING** |
| **Phishing Probability:** | **89.0%** | **Model Confidence:** | high |
| **Top Text Features:** | `httpurl,■ verify,■ password,■ update,■ account` | **Signal Role:** | Auxiliary ML signal (Non-overriding) |

## 6. AI Analyst Assistant Synthesis

| | | | |
|---|---|---|---|
| **Likely Attack Type:** | **Credential Harvesting /■ Brand Impersonation Spear-■Phishing** | **Likely Objective:** | Exfiltrate corporate Microsoft c■redentials via disguised credent■ial-■collection landing pages.■ |

**Executive Synthesis:** Suspicious message exhibiting critical credential-harvesting indicators with severe sender spoofing, including failed DMARC and SPF checks alongside a deceptive lookalike domain.

**Key Forensic Evidence Identified by LLM:**
- Failed DMARC (p=reject) and SPF authentication from sending IP 203.0.113.25
- Lookalike domain micros0ft-example.com imitating Microsoft (registered 18 days ago)
- HTML display anchor showing microsoft.com while pointing to evil-tracker.example
- NLP Phishing Classifier flagged high confidence (89% probability) for credential harvesting language
- Suspicious embedded shortened link (bit.ly/3sample-invoice)

## 7. Extracted Indicators of Compromise (IOCs)

### Observed URLs & Suspicion Scoring:

| EXTRACTED URL / TARGET HOST | SCORE | FORENSIC OBSERVATIONS & FLAGS |
|---|---|---|
| `https:■/■/■micros0ft-■example.■com/■login` | **25/100** | Potential brand impersonation:■ similar to microsoft.■com (91%■ match),■ Suspicious security/■authentication keywords found:■ login |
| `http:■/■/■evil-■tracker.■example/■login` | **40/100** | HTML display link mismatch:■ visible text claims &■#■x27;■microsoft.■com&■#■x27;■ but destination links to &■#■x27;■evil-■tracker.■example&■#■x27;■,■ Suspicious security/■authentication keywords found:■ login |
| `https:■/■/■bit.■ly/■3sample-■invoice` | **20/100** | URL shortener domain detected (b■it.■ly),■ Suspicious security/■authentication keywords found:■ invoice |

### Observed Attachments & Cryptographic Checksums:

| FILENAME | MIME TYPE | SIZE | SHA-256 CHECKSUM |
|---|---|---|---|
| **invoice_■2026.■pdf.■exe** | application/■octet-■stream | 145.0 KB | 9f86d081884c7d659a2feaa0■c55ad015a3bf4f1b2b0b822c■d15d6c15b0f00a08 |
| **terms_■and_■conditions.■pdf** | application/■pdf | 82.2 KB | 4a5e1e4baab89f3a32518a88■c31bc87f618f76673e2cc77a■b2127b7afdeda33b |

## 8. Network Transmission Path & Topology

| | | | |
|---|---|---|---|
| **Earliest Observable Sending IP:** | **198.■51.■100.■12** | **Reverse DNS / Host:** | smtp.■example.■org |

### Observed Infrastructure Route (Chronological Transmission Sequence):

| HOP | FROM (HOST / IP) | BY (RECEIVING RELAY) | TIMESTAMP |
|---|---|---|---|
| **#1** | smtp.■example.■org (198.■51.■100.■12) | mail.■example.■net (no IP) | Sun,■ 7 Sep 2026 10:■41:■59 +■0530 |
| **#2** | mail.■example.■net (203.■0.■113.■25) | mx.■company.■com (no IP) | Sun,■ 7 Sep 2026 10:■42:■01 +■0530 |

*Notice: Labeled strictly as observed infrastructure route. Does not represent physical attacker travel path.*

## 9. Infrastructure & Domain Intelligence

| IP ADDRESS | LOCATION (CITY/COUNTRY) | ASN / NETWORK | ORGANIZATION / ISP |
|---|---|---|---|
| 203.■0.■113.■25 | Amsterdam,■ Netherlands | AS12345 | Example Cloud Services |
| 198.■51.■100.■12 | Amsterdam,■ Netherlands | AS12345 | Example Cloud Services |

## 10. Brand Impersonation & Lookalike Domains

| SUSPECTED BRAND | SPOOFED DOMAIN | SIMILARITY | TECHNIQUES & ATTACK PROFILE |
|---|---|---|---|
| **Microsoft** | micros0ft-■example.■com | **91.0%** | character_■substitution,■ brand_■keyword |

## 11. Prescriptive Recommendations & Containment Actions

**[ ]** 1. Block sending IP 203.0.113.25 and domain micros0ft-example.com on perimeter email gateway
**[ ]** 2. Revoke active sessions and force password resets for any recipient who accessed the link
**[ ]** 3. Search SIEM proxy/DNS telemetry for outbound HTTP requests to evil-tracker.example and bit.ly/3sample-invoice
**[ ]** 4. Purge matching message IDs from all internal Exchange/M365 mailboxes

## 12. Evidentiary Limitations & Chain-of-Custody Certification

**Forensic Chain-of-Custody Certification:**
This document is generated by MailTrace AI based on literal RFC-822 evidence headers. Public IP geolocation represents coarse regional estimates based on RIR allocations and does not establish street-level physical locations or the personal identity of the sender. All cryptographic hash sums (SHA-256) were computed over unaltered message byte streams and preserve complete digital non-repudiation chain of custody.

| | | | |
|---|---|---|---|
| **EVIDENTIARY DIGITAL SEAL:** | SHA256:97d4b2e811c7520e5e79603f9050■d268159b360b9432df03d4083d8e■57ef228a | **VERIFICATION STATUS:** | **CRYPTOGRAPHICALLY VERIFIED** |